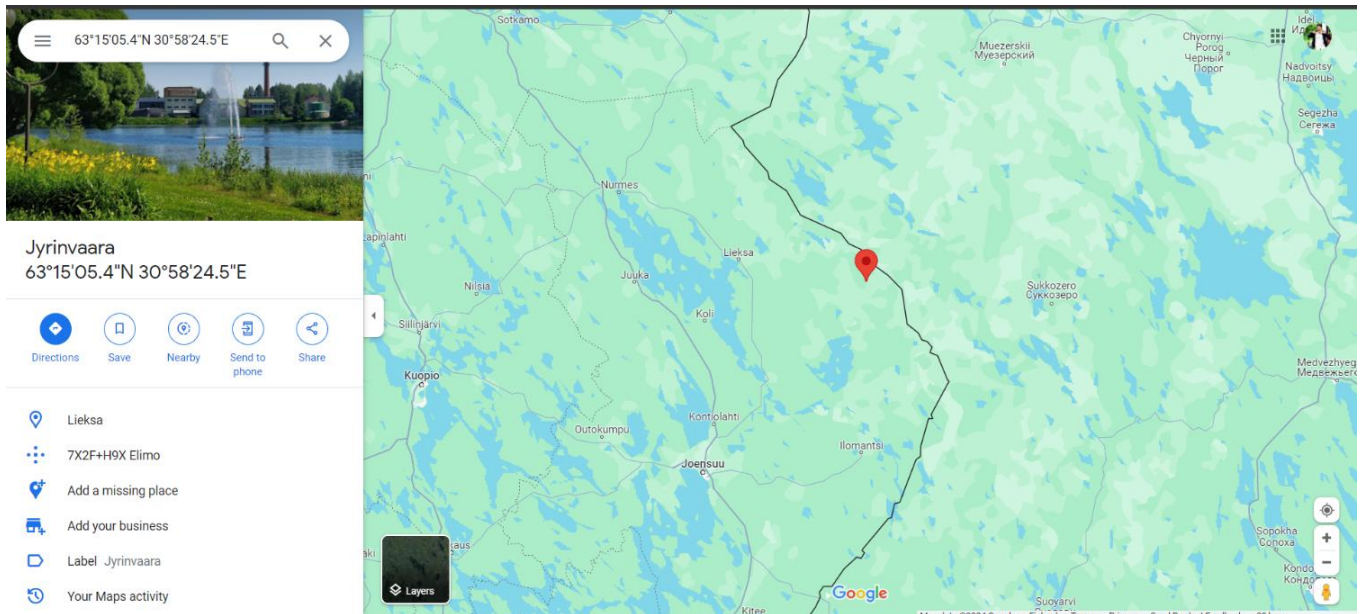


TREE SPECIES CLASSIFICATION WITH A DRONE IMAGE POINT CLOUD

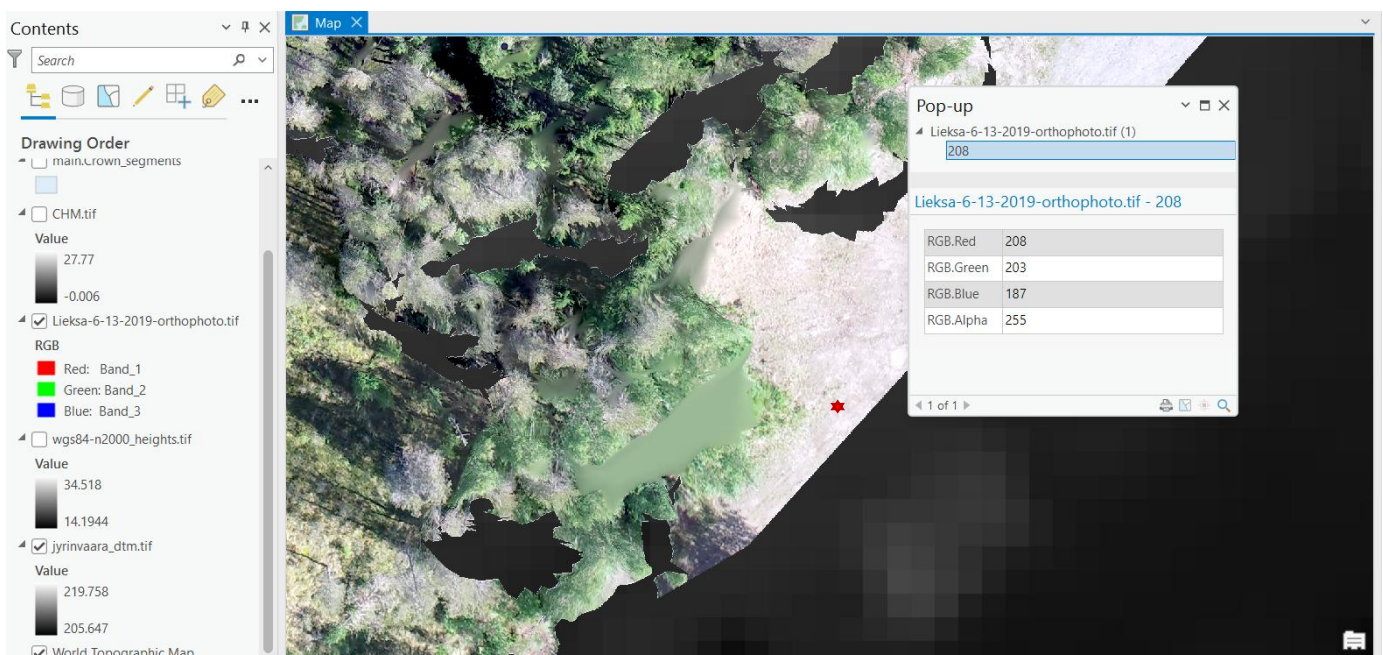
WeM lab model solution

1. A screenshot showing the location of Jyrinvaara within the region of Pohjois-Karjala on Google Maps (1 p):

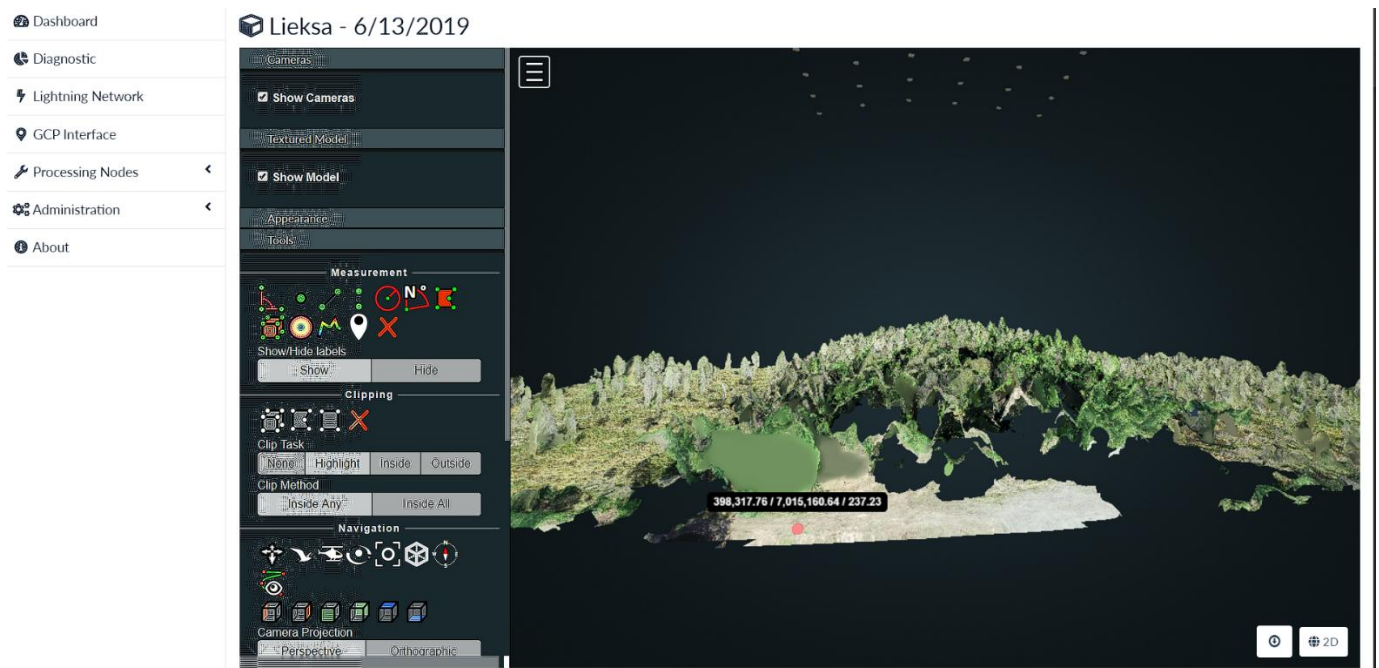


2. A comparison of theoretical and empirical elevation corrections for the point cloud and how the empirical correction was calculated. (1 p):

DTM elevation of the road: 208.1m

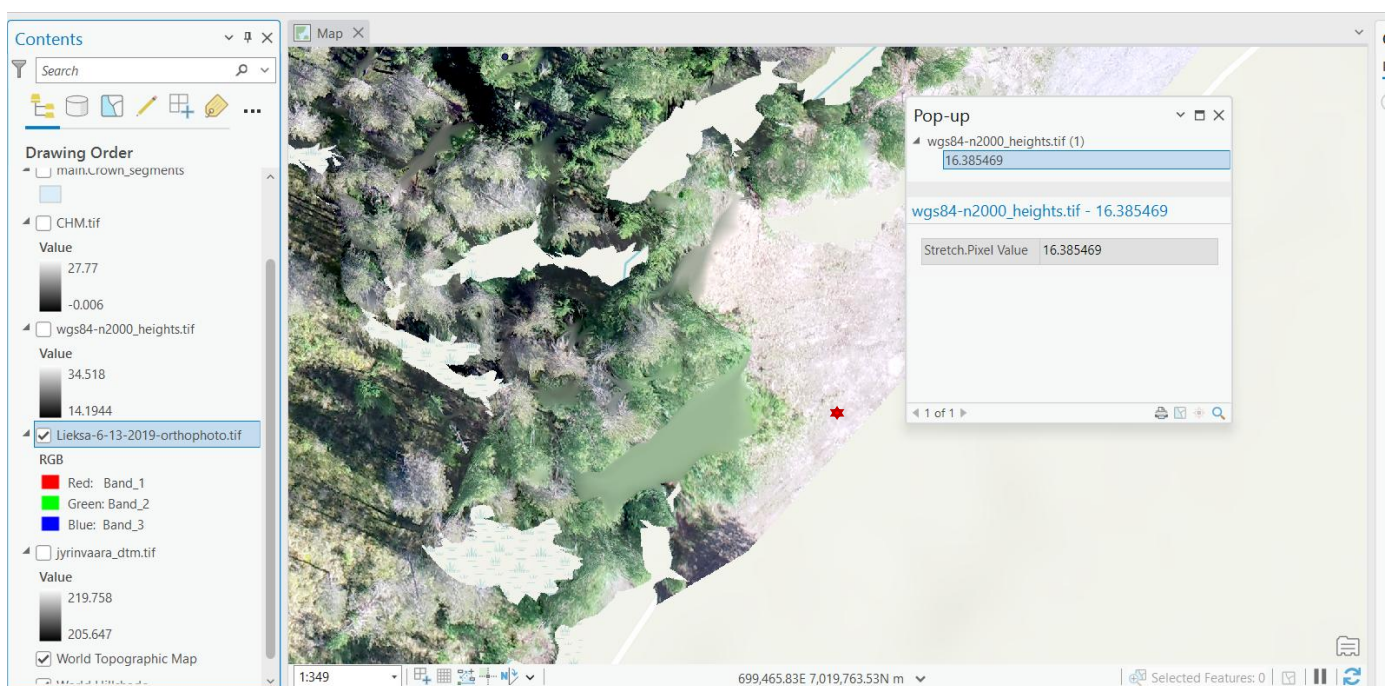


Point cloud elevation of the road: 237.2m

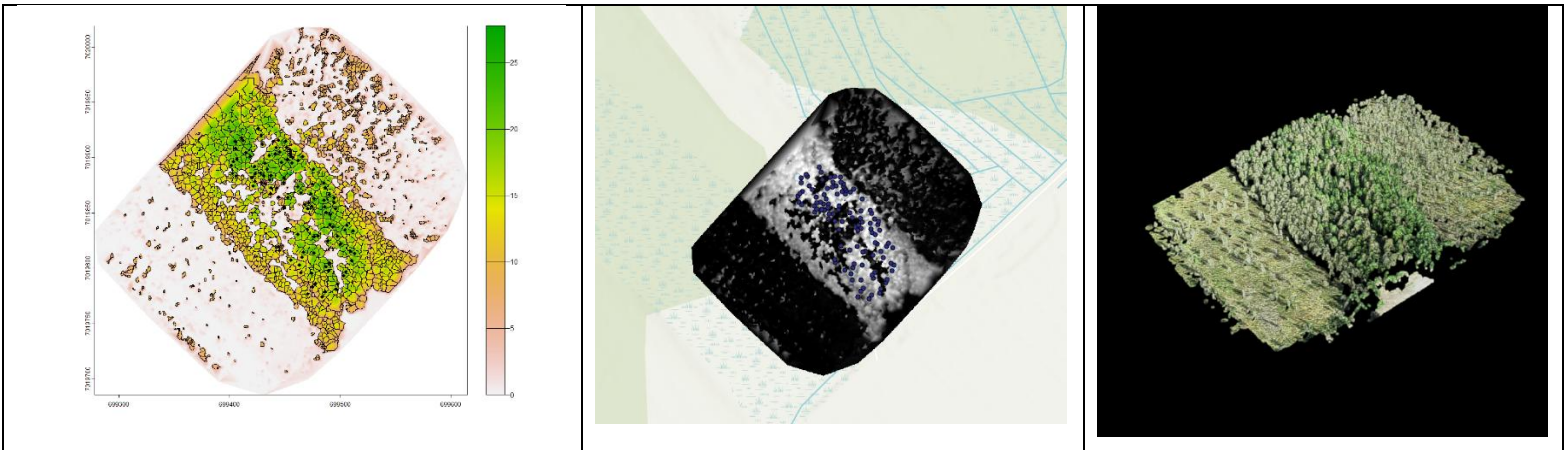


Calculate local correction factor: $237.2\text{m} - 208.1\text{m} = 29.1\text{m}$

Theoretical correction factor from the raster: 16.38m



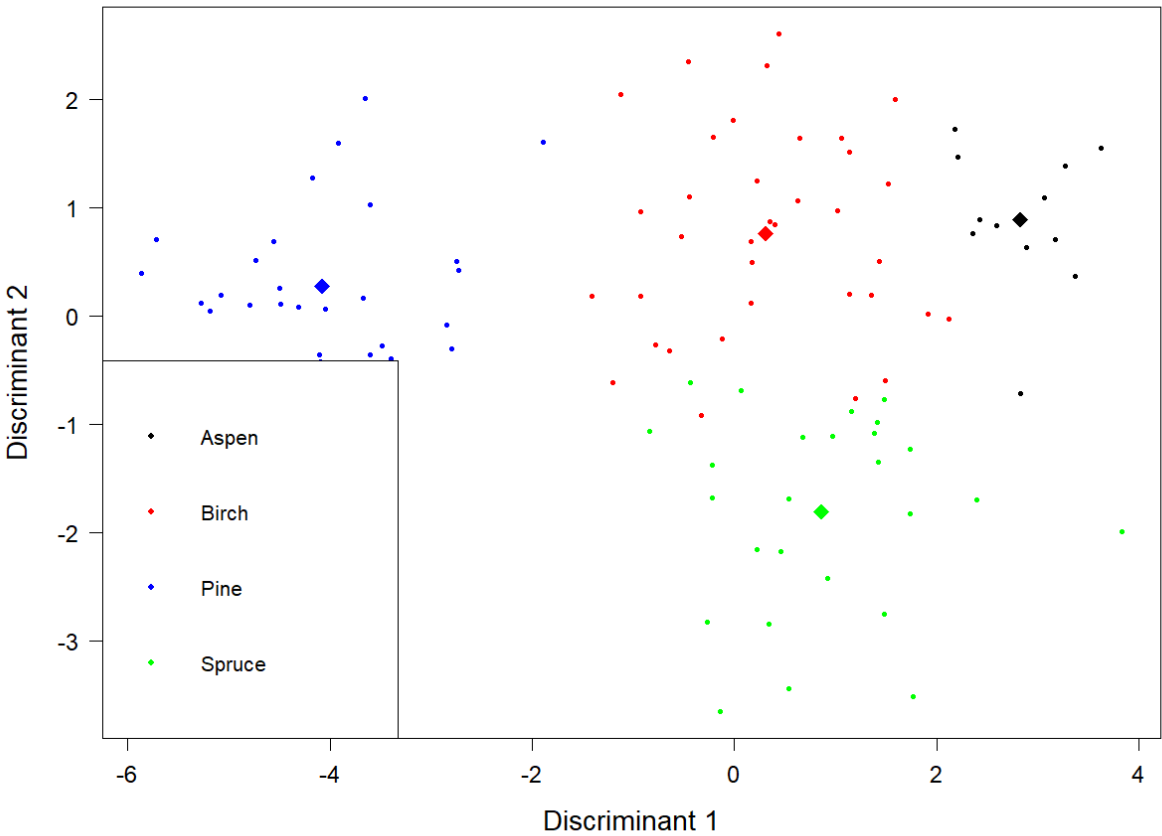
3. Screenshots comparing the crown segmentation results with the CHM and orthomosaic, and your own evaluation of the segmentation result based on visual interpretation of the screenshots.
(1 p)



The crown segmentation on the CHM distinctly outlines the tree crowns and also makes a clear distinction between the individual trees. The overlap between the neighbouring crowns is also minimal. The segmentation results show good agreement with the orthomosaic's visual cues, indicating precise tree crown demarcation. Though I believe it is difficult to properly comment on the segmentation precision in the absence of ground-truthing, the visual match between the segmented crowns and the actual canopy structure suggests a high degree of accuracy.

The 3D view from the orthomosaic clearly demarcates the fact that the crowns are distinctly elevated from the ground, and the variability in height shows the distinct ability between various species, which eventually iterates the fact that the forest is heterogeneous in composition.

4. Discriminant plot from LDA. (1 p)



The above discriminant plot iterates the fact that Discriminant 1 separates Pine from Birch, Aspen and Spruce. Likewise, Discriminant 2 separates Spruce from Birch, Aspen and Pine.

Additionally, the output (depicted below) from the `trainLDA()` function shows the proportion of the trace that Discriminant 1 explains 85% of the classification accuracy; Discriminant 2 explains 14%, and the remaining 1% by Discriminant 3.

```
> # View lda results
> lda$lda
Call:
lda(form, data = d, prior = pri)

Prior probabilities of groups:
  Aspen  Birch  Pine Spruce
  0.25   0.25   0.25   0.25

Group means:
      rel_h40      Rmean      R80      G80      B5
Aspen  0.82711142 -0.3848110 -0.7505148  1.1153548 -1.1045340
Birch  0.25999837 -0.2391148 -0.3189300  0.4140255 -0.1418397
Pine   0.09468179  1.0364896  1.1729747 -1.3618597  1.1139145
Spruce -0.92879369 -0.6913841 -0.5504033  0.4496137 -0.5750436

Coefficients of linear discriminants:
      LD1      LD2      LD3
rel_h40 -0.07323517  0.8992263  0.3062140
Rmean   1.13706988  1.0520774  0.3658877
R80     -1.81885265  0.4882498 -1.1230389
G80     1.17897954  2.4117712 -2.4179406
B5      -0.86289350  1.6327971 -1.9256191

Proportion of trace:
      LD1      LD2      LD3
0.8468  0.1417  0.0116
```

5. Error matrix from LDA and overall accuracy from it. (1 p)

```
> # View error matrix
> e.matrix <- lda$errmat
> e.matrix

      Aspen Birch Pine Spruce
Aspen    11     1    0     0
Birch     1    32    0     4
Pine      0     1   28     0
Spruce    0     2    0    22
> sum(diag(e.matrix))/sum(e.matrix)
[1] 0.9117647
```

The above figure shows the error matrix obtained and the accuracy predicted from it, which is about 91%.

6. First two discriminant functions as equations. (1 p)

LD1 = -0.07323517 * rel_h40 + 1.13706988 * Rmean - 1.81885265 * G95 + 1.17897954 * G80 - 0.86289350* B5

$$LD2 = 0.8992263 * rel_h40 + 1.0520774 * Rmean + 0.4882498 * G95 + 2.4117712 * G80 + 1.6327971 * B5$$

7. Interpretation of variable contributions to classification result. (2 p)

The model training initially started with a number of predictors as 2 and finally stopped when the number of predictors became 5, as there was no significant change in the overall accuracy when it became 6. The accuracy obtained while training with different predictors is shown in the table below.

Number of predictors	Accuracy
2	0.7647059
3	0.8333333
4	0.8823529
5	0.9117647
6	0.9313725

Coefficients of linear discriminants:

	LD1	LD2	LD3
rel_h40	-0.07323517	0.8992263	0.3062140
Rmean	1.13706988	1.0520774	0.3658877
R80	-1.81885265	0.4882498	-1.1230389
G80	1.17897954	2.4117712	-2.4179406
B5	-0.86289350	1.6327971	-1.9256191

In the finalised model of 5 predictors, R80 has the largest contribution to LD1. It has a negative coefficient (-1.81), so larger R80 values move the classification along the negative X-axis towards Pine and small values oppositely towards the rest of the species. Likewise, G80 has the largest coefficient in Discriminant 2 and is positive (2.41). Thus, large values for G80 move the prediction towards Aspen, Birch and Pine in the positive Y-axis and smaller values towards Spruce.